Answer key

Practice Test 7 (National level, hard) · 50 questions · 36 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	B	C	B	B	D	A	B	B	A	D	C	A	C

15
A

Section I: Multiple choice

1. A. The Mathematical Analysis of Logic

1. 1847: The Mathematical Analysis of Logic (a short book, written in a few weeks).
2. Laws of Thought is the bigger 1854 book (trap). The relay paper is Shannon's (1937); Die galvanische Kette is Ohm's (1827).

Note: A1 History/Boole
App: no app: fact or concept

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2. B. Electrophorus

1. Volta named it the "perpetual electrophorus" (1775). Wilcke had described a similar device in 1762.
2. The voltaic pile came later (1800); the torsion balance is Coulomb's.

Note: A1 History/Volta
App: no app: fact or concept

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3. C. Stays on its outer surface

1. Fourth memoir (1786): charge divides between touching conductors and stays on the outer surface.
2. He also found that charge leaks away slowly (through humid air and the supports), not at once.

Note: A1 History/Coulomb
App: no app: fact or concept

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4. B. James Clerk Maxwell

1. Maxwell (1860s) predicted EM waves moving at the speed of light.
2. Hertz made and detected them, and showed they reflect, refract and polarize like light.

Note: A1 History/Hertz
App: no app: fact or concept

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5. B. They begin on positive charges, end on negative charges, never cross, and are closer together where the field is stronger

1. Field lines show the direction of the force on a + test charge: out of +, into −.
2. Two lines cannot cross: the field has only one direction at each point.
3. Line density shows field strength.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: no app: fact or concept

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6. D. 8F

1. F is proportional to q_1q_2/r^2.
2. Doubling one charge: ×2. Half the distance: $\times(1/(1/2)^2) = \times4$. Total ×8.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: no app: fact or concept

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7. A. A shock then cannot take a path from hand to hand across the chest and heart

1. The most dangerous path is hand to hand or hand to foot through the chest.
2. With one hand away, a current path through the heart is much less likely.

Note: A3 DC and AC/Shock and Safety
App: no app: fact or concept

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8. B. Thermocouple

1. Two different metals joined at a hot and a cold junction give a small DC voltage (Seebeck effect).
2. Solar cell: light. Piezo: pressure. Fuel cell: a chemical reaction.

Note: A3 DC and AC/Other DC Sources
App: no app: fact or concept

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9. B. 11 V

1. V = IR = 2.5e-3 × 4.20e3 = 10.5 V on the calculator.
2. Multiplication: keep the fewest significant figures in the data. 2.5 mA has 2, so 11 V.

Note: A4 Current Voltage Resistance Power Energy/Units Prefixes and Sig Figs
App: `o i2.5m r4.2k` → `v Voltage = 10.5 V` *(checked)*

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10. A. All three bulbs go out, because the wire shorts the battery and a very large current flows (the fuse blows)

1. A wire across one parallel branch is a wire across the battery itself.
2. The voltage across every branch falls to about 0: all bulbs go out, and the large current blows the fuse.

Note: A5 DC Circuits/Open and Short Circuits
App: no app: fact or concept

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11. D. When the coil has no current

1. "Normal" = the coil is not energized. NC is closed to COM at rest.
2. When the coil is energized, COM moves to NO (normally open).

Note: A6 Switches Relays Fuses Breakers GFCI/Relays
App: no app: fact or concept

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12. C. The wire can carry 15-20 A and overheat (a fire risk) without the breaker tripping

1. A breaker protects the WIRE. Its rating must not be above the wire's rating.
2. 14 AWG needs a 15 A breaker; 20 A needs 12 AWG.

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: no app: fact or concept

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13. A. Keeps its voltage nearly constant while its current changes a lot

1. In breakdown the I-V curve is almost vertical: the voltage stays near V_Z.
2. That is why it is used as a simple voltage regulator.

Note: A7 Diodes and LEDs/Special Diodes
App: no app: fact or concept

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14. C. 10.6 V

1. In a bridge, the current goes through TWO diodes in each half cycle.
2. Peak out = 12.0 – 2 × 0.7 = 10.6 V. (11.3 V would be a single-diode rectifier.)

Note: A7 Diodes and LEDs/Diode Applications
App: no app: fact or concept

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15. A. {AND, OR}

1. Without NOT you cannot make an output 1 when all inputs are 0, so {AND, OR} cannot make NOT.
2. NAND alone and NOR alone are universal. AND + NOT gives NAND, so it is complete too.

Note: A8 Digital Logic/Universal Gates (NAND and NOR)
App: no app: fact or concept

Section II: Short answer and calculations

16. 1.7 N, along the diagonal, away from the center of the square

1. Each neighbor (0.10 m away): F = 8.99e9 × (1.0e-6)^2 / 0.10^2 = 0.8990 N.
2. The two neighbor forces are perpendicular: together √2 × 0.8990 = 1.2714 N, along the diagonal.
3. The far corner is at a distance with r^2 = 2 × 0.10^2 = 0.020 m^2: F = 0.4495 N, along the same diagonal.
4. Total = 1.2714 + 0.4495 = 1.721 N = 1.7 N, pointing away from the center.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 1u 0 0, 1u .1 0, 1u 0 .1, 1u .1 .1` → `F on q4: 1.72 N at 45 deg` *(checked)*
App: `Q 1u 0 0, 1u .1 0, 1u 0 .1, 1u .1 .1` → `F on q4: 1.72 N at 45 deg` *(checked)*

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17. U = -0.22 J; no: the work done to assemble them is negative (the charges give energy)

1. U = k(q_1q_2/r_{12} + q_1q_3/r_{13} + q_2q_3/r_{23}), with signs. r_{23} = 0.50 m (3-4-5 triangle).
2. q_1q_2/0.30 = -2.000e-11; q_1q_3/0.40 = 2.000e-11; q_2q_3/0.50 = -2.400e-11 (C^2/m).
3. Sum = -2.400e-11; × 8.99e9 = -0.2158 J = -0.22 J.
4. Work to assemble = U. It is negative, so the group pulls itself together (energy is given out).

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `Q 2u 0 0, −3u .3 0, 4u 0 .4` → `Total U (all pairs): −215.7 mJ` *(checked)*

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18. +1.6 nC

1. Losing electrons adds positive charge: 3.0e10 × 1.602e-19 = 4.806 nC.
2. New charge = –3.2 + 4.806 = +1.606 nC.
3. Addition: keep the tenths place (from –3.2): +1.6 nC.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `a e3*10^10` → `q Charge = 4.807 nC` *(checked)*

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19. (a) 2.14 A; (b) 1.36 A

1. (a) Series: EMF 3.00 V, internal 0.400 Ω. I = 3.00 / (0.400 + 1.00) = 2.14 A.
2. (b) Parallel: EMF 1.50 V, internal 0.200/2 = 0.100 Ω. I = 1.50 / (0.100 + 1.00) = 1.36 A.
3. Series gives more current here because the load is much larger than r.

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `i e3 r.4 l1` → `i Current = 2.143 A` *(checked)*
App: `i e1.5 r.1 l1` → `i Current = 1.364 A` *(checked)*

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20. 36 h

1. Energy = 4.000 Ah × 3.80 V = 15.20 Wh.
2. t = E/P = 15.20 / 0.42 = 36.2 h = 36 h (0.42 W has 2 significant figures).

Note: A3 DC and AC/Batteries and Cells
App: `b c4000 v3.8 p.42` → `t Time (hours) = 36.19 h` *(checked)*

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21. (a) 20.0 kW; (b) 200 W

1. (a) I = P/V = 1.00e6 / 10.0e3 = 100.0 A. Loss = $I^2R = 100^2 \times 2.00$ = 20.0 kW.
2. (b) I = 1.00e6 / 100e3 = 10.0 A. Loss = $10^2 \times 2.00$ = 200 W.
3. Ten times the voltage: one tenth the current, one hundredth the loss.

Note: A3 DC and AC/AC Characteristics
App: `o i100 r2` → `p Power = 20 kW` *(checked)*
App: `o i10 r2` → `p Power = 200 W` *(checked)*

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22. 120 mA: in the 100-200 mA range that can cause ventricular fibrillation

1. I = 120 / 1000 = 120 mA.
2. About 1 mA can be felt, 10-20 mA stops you letting go, and about 100-200 mA across the chest can make the heart fibrillate (stop pumping).

Note: A3 DC and AC/Shock and Safety
App: `o v120 r1k` → `i Current = 120 mA` *(checked)*

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23. 5.09 Ω; 76.3 V

1. A = $\pi d^2/4 = \pi \times (2.05\text{e-}3)^2/4$ = 3.3006e-06 m^2.
2. R = ρL/A = 1.68e-8 × 1000 / 3.3006e-06 = 5.09 Ω.
3. V = IR = 15.0 × 5.0899 = 76.3 V.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `r p1.68*10^-8 l1000 d2.05m` → `r Resistance = 5.09 ohm` *(checked)*
App: `o i15 r5.0899` → `v Voltage = 76.35 V` *(checked)*

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24. 2.0×10^5 C; 2.4×10^6 J

1. 1 Ah = 1 A × 3600 s = 3600 C. Q = 55 × 3600 = 198000 C = 2.0×10^5 C (55 has 2 significant figures).
2. E = QV = 198000 × 12.0 = 2.376e+06 J = 2.4×10^6 J.

Note: A4 Current Voltage Resistance Power Energy/Current and Charge
App: `b c55Ah v12` → `q Charge = 198 kC` *(checked)*
App: `b c55Ah v12` → `j Energy (joules) = 2.376 MJ` *(checked)*

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25. (a) 100 kΩ and 22 Ω; (b) orange, orange, black, brown, green

1. (a) 1003: digits 100, then 3 zeros: 100 000 ohm. 220: digits 22, then 0 zeros: 22 ohm (not 220!).
2. (b) 3, 3, 0 = orange, orange, black. 330 × 10 = 3300: multiplier brown. ±0.5% = green.
3. The app shows the ±1% band (brown) last; for ±0.5% write green.

Note: A5 DC Circuits/Resistor Color Codes
App: `c1003` → `If SMD code: 1003 = 100 kohm` *(checked)*
App: `c220` → `If SMD code: 220 = 22 ohm` *(checked)*
App: `c3.3k` → `5-band: orange orange black brown brown` *(checked)*

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26. 2.50 kΩ

1. The ammeter measures the current of the resistor AND the voltmeter.
2. Voltmeter current: 4.80 / 10.0k = 0.480 mA.
3. Resistor current: 2.40 – 0.480 = 1.92 mA.
4. R = 4.80 / 0.00192 = 2.50 kΩ (not 4.80/2.40 mA = 2.00 kΩ).

Note: A5 DC Circuits/Meters and Measurement
App: `o v4.8 r10k` → `i Current = 480 uA` *(checked)*
App: `o v4.8 i1.92m` → `r Resistance = 2.5 kohm` *(checked)*

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27. 15.0 Ω

1. Number the junctions 1-9 row by row; measure from 1 to 9.
2. By symmetry about the 1-9 diagonal, 2 and 4 are at the same voltage, 3 and 7 are too, and 6 and 8 are too. Join them.
3. 1 to {2,4}: two R in parallel = R/2. {6,8} to 9: R/2.
4. From {2,4} to {6,8} there are two paths: through 5 (R/2 + R/2 = R) and through {3,7} (R/2 + R/2 = R). In parallel: R/2.
5. Total = R/2 + R/2 + R/2 = 1.5R = 15.0 ohm.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1-2-3 10 4-5-6 7-8-9 1-4-7 2-5-8 3-6-9 =1-9` → `R(1-9) = 15 ohm` *(checked)*

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28. 23.3 A; 30 A breaker

1. I = P/V = 5600 / 240 = 23.3 A.
2. Breaker needed: 23.33 / 0.80 = 29.2 A.
3. Smallest standard size at or above that: 30 A.

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: `o p5.6k v240` → `i Current = 23.33 A` *(checked)*

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29. 11 A; 15 A fuse

1. Total power = 55 + 55 + 21 = 131 W.
2. I = 131 / 12.0 = 10.92 A = 11 A (55 W has 2 significant figures).
3. 125%: 10.92 × 1.25 = 13.6 A. Smallest size at or above that: 15 A.

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: `o p131 v12` → `i Current = 10.92 A` *(checked)*

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30. (a) 170 Ω; (b) 98.8 mW

1. Load current: 5.60 / 280 = 20.0 mA.
2. (a) The worst case is the LOWEST supply: the resistor must still pass the whole load current (zener current ≥ 0).
3. R_{max} = (9.00 − 5.60) / 0.0200 = 3.40 / 0.0200 = 170 Ω.
4. (b) At 12.00 V: I_R = 6.40 / 170 = 37.65 mA; zener takes 17.65 mA.
5. P_Z = 5.60 × 0.01765 = 98.8 mW.

Note: A7 Diodes and LEDs/Special Diodes
App: `N V1=0 9, 1=2 x, Z0=2 5.6, 2=0 280, V2=5.6` → `Any x up to 170 ohm` *(checked)*
App: `N V1=0 12, 1=2 170, Z0=2 5.6, 2=0 280` → `Z0=2 5.6: ON zener (breakdown) 17.65 mA 2>0 98.82 mW` *(checked)*

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31. F = B′D′ + BD: an XNOR of B and D

1. Corners 0, 2, 8, 10 (B = 0, D = 0): B'D'.
2. Use the don't care 15 as a 1: 5, 7, 13, 15 (B = 1, D = 1): BD.
3. F = B'D' + BD = (B ⊕ D)': an XNOR of B and D (A and C do not matter).

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t abcd m0,2,5,7,8,10,13 d15` → `F = B'D' + BD` *(checked)*

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32. 1000010 = 66

1. 10110 = 22, 11101 = 29, 1111 = 15.
2. 22 + 29 + 15 = 66.
3. 66 = 64 + 2 = 1000010.
4. Column by column, a column can add to 3: write 1 and carry 1 (and a sum of 4 or 5 carries 2 columns over).

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x10110+11101+1111` → `sum bin 1000010` *(checked)*

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33. (a) 1; (b) S = 00100000, C_{out} = 1

1. (a) The carry into bit 4 is the carry out of the low 4 bits: 0101 + 1011 = 5 + 11 = 16, which is 16 or more: carry = 1.
2. (b) 181 + 107 = 288 = 100100000 (9 bits).
3. S = the low 8 bits = 00100000; C_{out} = 1.

Note: A8 Digital Logic/Adders
App: `x10110101+01101011` → `sum bin 100100000` *(checked)*

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34. D_0..D_7 = D, 1, 0, D′, D, 0, 1, D′

1. Each select value ABC picks a pair of rows (D = 0, D = 1).
2. ABC = 000: rows 0, 1 = (0, 1): D. 001: rows 2, 3 = (1, 1): 1. 010: rows 4, 5 = (0, 0): 0. 011: rows 6, 7 = (1, 0): D'.
3. 100: rows 8, 9 = (0, 1): D. 101: rows 10, 11 = (0, 0): 0. 110: rows 12, 13 = (1, 1): 1. 111: rows 14, 15 = (1, 0): D'.

Note: A8 Digital Logic/Multiplexers and Decoders
App: `t abcd m1,2,3,6,9,12,13,14` → `F = A'B'C + ABC' + B'C'D + BCD'` *(checked)*
App: `t abcd m1,2,3,6,9,12,13,14` → `Mux 8:1, select A,B,C: D0.. = D 1 0 D' D 0 1 D'` *(checked)*

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35. (a) 15 W, 27 W, 45 W, 65 W; (b) 49.5 min

1. (a) P = VI: 5 × 3 = 15 W; 9 × 3 = 27 W; 15 × 3 = 45 W; 20 × 3.25 = 65 W.
2. (b) Energy into the battery: 80% of 57.0 Wh = 45.6 Wh. From the charger: 45.6 / 0.850 = 53.65 Wh.
3. t = 53.65 Wh / 65 W = 0.8253 h = 49.5 min.

Note: A9 Hands-On and Meters/Task - USB Charger
App: `z w45.6 g65 e.85` → `t Charge time = 0.82534 h` *(checked)*

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36. Both LEDs and the resistor in series; 280 Ω

1. In a series loop every part carries the same current, so one resistor is enough.
2. The resistor gets 9.00 – 1.80 – 3.00 = 4.20 V.
3. R = 4.20 / 0.0150 = 280 Ω.
4. (In parallel, each LED would need its own resistor, because their forward voltages differ.)

Note: A9 Hands-On and Meters/Task - Equal LEDs
App: `l s9 f4.8 i15m` → `r Resistor = 280 ohm` *(checked)*

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37. (1.00 k + 330) || (220 + 150) = 289.5 Ω

1. Two series pairs in parallel: 1000 + 330 = 1330 ohm and 220 + 150 = 370 ohm.
2. 1330 × 370 / 1700 = 289.47 ohm.
3. Error = -0.18%. The next best is (330 + 220) || (470 + 150) = 291.5 ohm (+0.50%).

Note: A9 Hands-On and Meters/Task - Target Resistance
App: `h 150 220 330 470 1k=290` → `289.5 (-0.183%) (1k+330)||(220+150)` *(checked)*
App: `s(1k+330)/(220+150)` → `Req = 289.5 ohm` *(checked)*

Section III: Circuit problems

Problem A: two batteries, two nodes (Kirchhoff's laws) (Questions 38–41)

E1 = 20.0 V feeds node A through R1 = 4.00 Ω. R2 = 10.0 Ω joins A to the bottom wire (ground). R3 = 5.00 Ω joins A to node B. R4 = 20.0 Ω joins B to ground. E2 = 8.00 V connects to B through R5 = 2.00 Ω. Both batteries have their + plate at the TOP. Let V_A and V_B be the node voltages above ground.

38. 12.2 V

1. Node A (currents out = 0): $(V_A - 20.0)/4.00 + V_A/10.0 + (V_A - V_B)/5.00 = 0$.
2. Node B: $(V_B - V_A)/5.00 + V_B/20.0 + (V_B - 8.00)/2.00 = 0$.
3. Times 20: 11 V_A – 4 V_B = 100 and –4 V_A + 15 V_B = 80.
4. Determinant 11 × 15 – 16 = 149. V_A = (1500 + 320)/149 = 1820/149 = 12.2 V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a11 b-4 c100 d-4 e15 f80` → `x x (I1 or V1) = 12.215` *(checked)*
App: `N V1-0 20, 1-2 4, 2-0 10, 2-3 5, 3-0 20, V4-0 8, 4-3 2` → `V1 = 20 V V2 = 12.21 V` *(checked)*

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39. 8.59 V

1. V_B = (11 × 80 + 4 × 100)/149 = 1280/149 = 8.59 V.
2. Check node B: (8.5906 – 12.2148)/5 + 8.5906/20 + (8.5906 – 8)/2 = 5.6e-17 (zero).

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a11 b-4 c100 d-4 e15 f80` → `y y (I2 or V2) = 8.5906` *(checked)*
App: `N V1-0 20, 1-2 4, 2-0 10, 2-3 5, 3-0 20, V4-0 8, 4-3 2` → `V3 = 8.591 V V4 = 8 V` *(checked)*

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40. 725 mA from A to B

1. I = (V_A – V_B)/5.00 = (540/149)/5.00 = 725 mA.
2. Use the exact fractions (or 4+ digits): subtracting rounded 12.2 and 8.59 would lose digits.
3. V_A is higher, so the current flows from A to B.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 20, 1–2 4, 2–0 10, 2–3 5, 3–0 20, V4–0 8, 4–3 2` → `R2–3 5: 3.624 V 724.8 mA 2>3 2.627 W` *(checked)*

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41. Being charged: 295 mA flows INTO its + plate; it takes in 2.36 W

1. Current in R5 from B toward E2: (V_B – 8.00)/2.00 = (88/149)/2.00 = 295 mA.
2. V_B is above 8.00 V, so the current enters E2 at its + plate: E2 is being charged.
3. P = 8.00 × 0.2953 = 2.36 W taken in.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 20, 1–2 4, 2–0 10, 2–3 5, 3–0 20, V4–0 8, 4–3 2` → `V4–0 8: 295.3 mA into +, takes 2.362 W` *(checked)*
App: `N V1–0 20, 1–2 4, 2–0 10, 2–3 5, 3–0 20, V4–0 8, 4–3 2` → `V4–0 8: 295.3 mA into +, takes 2.362 W` *(checked)*

Problem B: a bridge that is not balanced (Questions 42–45)

A 12.0 V ideal battery feeds the bridge below. P is between R1 = 10.0 Ω (top) and R2 = 20.0 Ω (bottom). Q is between R3 = 30.0 Ω (top) and R4 = 10.0 Ω (bottom). R5 = 25.0 Ω joins P and Q.

42. 5.00 V

1. V_P = 12.0 × 20.0/30.0 = 8.00 V.
2. V_Q = 12.0 × 10.0/40.0 = 3.00 V.
3. V_P – V_Q = 5.00 V (P is higher).

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1–0 12 1–2 10 2–0 20 1–3 30 3–0 10 V2–3` → `V2 – V3 = 5 V` *(checked)*

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43. 0.128 A from P to Q

1. Node P: (V_P – 12)/10 + V_P/20 + (V_P – V_Q)/25 = 0.
2. Node Q: (V_Q – 12)/30 + V_Q/10 + (V_Q – V_P)/25 = 0.
3. Solve: V_P = 7.1489 V, V_Q = 3.9574 V.
4. I = (7.1489 – 3.9574)/25 = 128 mA, from P to Q.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 12 1–2 10 2–0 20 1–3 30 3–0 10 2–3 25` → `R2–3 25: 3.191 V 127.7 mA 2>3 407.4 mW` *(checked)*

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44. 15.9 Ω

1. Battery current = (12 – V_P)/10 + (12 – V_Q)/30 = 0.4851 + 0.2681 = 0.7532 A.
2. R_{eq} = 12.0 / 0.7532 = 15.9 Ω.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1–2 10 2–0 20 1–3 30 3–0 10 2–3 25 =1–0` → `R(1–0) = 15.93 ohm` *(checked)*
App: `N V1–0 12 1–2 10 2–0 20 1–3 30 3–0 10 2–3 25` → `V1–0 12: 753.2 mA out of +, gives 9.038 W` *(checked)*

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45. 847 mA

1. Now P and Q are one node: R1 || R3 on top, R2 || R4 on the bottom, in series.
2. 10.0 || 30.0 = 7.50 ohm. 20.0 || 10.0 = 6.667 ohm. Total = 14.167 ohm.
3. I = 12.0 / 14.167 = 847 mA.

Note: A5 DC Circuits/Open and Short Circuits
App: `N V1-0 12 1-2 10 2-0 20 1-2 30 2-0 10` → `V1-0 12: 847.1 mA out of +, gives 10.16 W` *(checked)*
App: `s10/30+20/10` → `Req = 14.17 ohm` *(checked)*

Problem C: NAND-only logic (Questions 46–48)

Four NAND gates: G1 = NAND(A, B), G2 = NAND(C, G1), G3 = NAND(G1, D), and F = G4 = NAND(G2, G3).

46. F = (AB)′(C + D) = A′C + A′D + B′C + B′D

1. F = (G2 · G3)' = G2' + G3' (De Morgan).
2. G2' = G1 · C and G3' = G1 · D, so F = G1(C + D).
3. G1 = (AB)' = A' + B'. F = (A' + B')(C + D) = A'C + A'D + B'C + B'D.
4. No term covers another, so this is the shortest SOP (4 terms).

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t-(-(c-(ab))-(-(ab)d))` → `F = A'C + A'D + B'C + B'D` *(checked)*

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47. 9 rows: m(1, 2, 3, 5, 6, 7, 9, 10, 11)

1. F = 1 needs (AB)' = 1 (AB = 00, 01 or 10: 3 ways) AND (C + D) = 1 (CD = 01, 10 or 11: 3 ways).
2. 3 × 3 = 9 rows.
3. AB = 00: 1, 2, 3. AB = 01: 5, 6, 7. AB = 10: 9, 10, 11.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t-(-(c-(ab))-(-(ab)d))` → `m(1,2,3,5,6,7,9,10,11)` *(checked)*

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48. F = (A′ + B′)(C + D)

1. From F = G1(C + D) with G1 = A' + B'.
2. Check: F = 0 when AB = 11 (A' + B' = 0) or when CD = 00 (C + D = 0): 4 + 4 − 1 = 7 rows, and 16 − 7 = 9.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t(-a+-b)(c+d)` → `F = A'C + A'D + B'C + B'D` *(checked)*

Problem D: zener and diode (Questions 49–50)

+12.00 V feeds R1 = 1.00 kΩ to node X. A 5.10 V zener diode goes from X to ground (cathode at X). A silicon diode D1 (0.70 V when on, anode at X) goes from X to node Y. R2 = 2.20 kΩ joins Y to ground.

49. V_X = 5.10 V; V_Y = 4.40 V

1. Guess: the zener is in breakdown, so V_X = 5.10 V, and D1 is on, so V_Y = 5.10 − 0.70 = 4.40 V.
2. Check: R1 brings (12.00 − 5.10)/1000 = 6.90 mA; D1 and R2 take 4.40/2200 = 2.00 mA.
3. The rest, 4.90 mA, flows in the zener in the reverse direction (positive): the guess is right.
4. (Without the zener, X would be at about 0.7 + 11.3 × 2.2/3.2 = 8.5 V, more than 5.10 V.)

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 12, 1-2 1k, Z0-2 5.1, D2-3, 3-0 2.2k` → `V3 = 4.4 V` *(checked)*
App: `N V1-0 12, 1-2 1k, Z0-2 5.1, D2-3, 3-0 2.2k` → `Z0-2 5.1: ON zener (breakdown) 4.9 mA 2>0 24.99 mW` *(checked)*

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50. 4.90 mA; 25.0 mW

1. I_Z = 6.90 − 2.00 = 4.90 mA (subtraction keeps the 0.01 mA place: 3 s.f.).
2. P = V_Z I_Z = 5.10 × 0.00490 = 25.0 mW.

Note: A7 Diodes and LEDs/Special Diodes
App: `N V1-0 12, 1-2 1k, Z0-2 5.1, D2-3, 3-0 2.2k` → `Z0-2 5.1: ON zener (breakdown) 4.9 mA 2>0 24.99 mW` *(checked)*
App: `N V1-0 12, 1-2 1k, Z0-2 5.1, D2-3, 3-0 2.2k` → `Z0-2 5.1: ON zener (breakdown) 4.9 mA 2>0 24.99 mW` *(checked)*
